

Exp. 6	CDN SERVICE SELECTION AND CONFIGURATION USING AMAZON CLOUDFRONT
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AIM

To select a Content Delivery Network (CDN) provider, configure a CDN distribution, set up DNS records, and deliver web content through the CDN network using Amazon CloudFront.

OBJECTIVES

- To understand the concept of Content Delivery Networks (CDN).
- To study CDN providers and their features.
- To configure Amazon CloudFront as a CDN service.
- To configure DNS records for CDN integration.
- To create and manage CDN distributions.
- To verify content delivery through CDN edge locations.

SOFTWARE REQUIREMENTS

- AWS Account
- Amazon S3
- Amazon CloudFront
- Domain Name (Optional)
- Web Browser
- Internet Connection

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THEORY

A Content Delivery Network (CDN) is a distributed network of servers that delivers web content from locations closest to users. CDN services reduce latency, improve website performance, enhance availability, and provide security features

Popular CDN providers include:

- Cloudflare
- Amazon CloudFront
- Akamai
- Microsoft Azure CDN
- Google Cloud CDN

Amazon CloudFront is AWS's CDN service that distributes content globally through edge locations. It integrates seamlessly with Amazon S3, EC2, Load Balancers, and custom origins.

DNS (Domain Name System) configuration is essential in CDN deployment. DNS records map domain names to CDN distributions, allowing users to access content through custom domains.

CDN PROVIDER COMPARISON

Provider	Features
Cloudflare	Free plan, DNS management, DDoS protection
Amazon CloudFront	Global AWS integration, scalable CDN
Akamai	Enterprise-grade CDN services
Azure CDN	Microsoft ecosystem integration
Google Cloud CDN	Integrated with Google Cloud

ARCHITECTURE

User Browser



DNS Resolution



Amazon CloudFront Distribution



Edge Location



Amazon S3 Bucket (Origin)



Content Delivery

ALGORITHM

1. Create AWS account.
2. Create S3 bucket.
3. Upload website content.
4. Configure S3 permissions.
5. Create CloudFront distribution.
6. Select S3 bucket as origin.
7. Configure CDN settings.
8. Configure DNS records.
9. Access content through CloudFront URL.
10. Verify CDN delivery.

FLOWCHART

Start



Create S3 Bucket



Upload Content



Create CloudFront Distribution



Configure Origin



Configure DNS



Deploy Distribution



Access CDN URL



Verify Content Delivery



Stop

PROCEDURE

STEP 1: LOGIN TO AWS CONSOLE

1. Open AWS Management Console.
2. Login using AWS credentials.
3. Navigate to Amazon S3.

STEP 2: CREATE S3 BUCKET

1. Click **Create Bucket**.
2. Enter bucket name:

cdn-lab-bucket

3. Select Region.
4. Click **Create Bucket**.

STEP 3: UPLOAD WEBSITE CONTENT

Create sample webpage:

```
<html>
<head>
<title>CDN Experiment</title>
</head>
<body>
<h1>Welcome to CloudFront CDN</h1>
</body>
</html>
```

Save as:

index.html

Upload file to S3 bucket.

STEP 4: ENABLE STATIC WEBSITE HOSTING

1. Open S3 Bucket.
2. Navigate to **Properties**.
3. Enable **Static Website Hosting**.
4. Set:

Index Document: index.html

5. Save changes.

STEP 5: CONFIGURE BUCKET PERMISSIONS

1. Navigate to **Permissions**.
2. Configure bucket policy to allow CloudFront access.
3. Save configuration.

STEP 6: CREATE CLOUDFRONT DISTRIBUTION

1. Open Amazon CloudFront Console.
2. Click **Create Distribution**.
3. Under Origin Domain select:

`cdn-lab-bucket.s3.amazonaws.com`

4. Configure default settings.
5. Enable caching.
6. Click **Create Distribution**.

STEP 7: WAIT FOR DISTRIBUTION

CloudFront creates a unique distribution URL:

Example:

`d123abcdxyz.cloudfront.net`

Deployment may take several minutes.

STEP 8: TEST CDN DISTRIBUTION

Open browser:

`https://d123abcdxyz.cloudfront.net`

Verify webpage loads successfully.

STEP 9: CONFIGURE CUSTOM DOMAIN (OPTIONAL)

Suppose domain:

`www.example.com`

Open DNS management panel.

STEP 10: VERIFY DNS RESOLUTION

Check DNS mapping:

```
nslookup www.example.com
```

OR

```
dig www.example.com
```

Verify CloudFront distribution is returned.

STEP 11: VERIFY CDN DELIVERY

Access:

```
https://www.example.com
```

Verify content is served successfully through CloudFront.

STEP 12: VIEW DISTRIBUTION STATISTICS

1. Open CloudFront Console.
2. Select Distribution.
3. View:

- Requests
- Data Transfer
- Cache Hit Ratio
- Geographic Distribution

COMMANDS USED

```
nslookup www.example.com
```

```
dig www.example.com
```

AWS Services Used:

- Amazon S3
- Amazon CloudFront

OBSERVATIONS

Parameter	Status
AWS Account Creation	Successful
S3 Bucket Creation	Successful
Website Upload	Successful
Static Hosting Enabled	Successful
CloudFront Distribution Creation	Successful
DNS Configuration	Successful
CDN URL Access	Successful
Content Delivery Verification	Successful

EXPECTED OUTPUT

- Website content uploaded successfully.
- CloudFront distribution created.
- DNS records configured.
- CDN URL accessible.
- Content delivered through CDN edge locations.
- Reduced latency and improved performance.

RESULT

A Content Delivery Network was successfully implemented using Amazon CloudFront. Website content was stored in Amazon S3, distributed through CloudFront edge locations, and accessed through CDN URLs and DNS configuration.